

CHICAGO LEWIS, IL, USA

WMO: 725348

Lat: 41.604N

Lon: 88.085W

Elev: 660

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.3	2.6	-10.8	3.2	-1.7	-5.9	4.1	4.1	28.5	30.7	26.2	29.6	9.7	300	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	90.7	74.0	88.3	72.9	85.8	71.9	78.0	86.5	76.1	84.2	74.4	82.0	11.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	135.8	82.9	73.2	126.6	80.8	72.1	121.8	79.6	41.9	87.0	40.0	83.9	38.3	82.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	23.0	19.8	-10.3	94.6	6.4	2.7	-14.8	96.5	-18.6	98.0	-22.2	99.5	-26.8	101.5
			-11.1	80.8	6.0	1.5	-15.4	81.8	-19.0	82.7	-22.4	83.6	-26.7	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	24.1	26.6	38.7	50.3	60.4	69.9	73.3	72.0	65.8	52.8	40.5	29.4
	DBStd	19.69	11.83	11.48	11.19	9.60	9.09	6.42	5.38	5.18	7.56	9.33	9.57	10.70
	HDD50	2999	803	657	383	111	14	0	0	0	1	77	310	643
	HDD65	6162	1267	1075	817	452	203	24	4	6	81	393	735	1105
	CDD50	3158	1	2	33	119	335	598	722	681	474	165	25	3
	CDD65	847	0	0	3	9	60	172	261	222	104	16	0	0
	CDH74	7507	0	0	20	128	595	1559	2373	1813	877	142	0	0
	CDH80	2404	0	0	1	26	184	521	810	562	268	32	0	0
Wind	WSAvg	9.3	10.8	10.5	10.8	11.4	9.9	8.5	7.0	6.3	7.2	8.9	10.2	10.2
Precipitation	PrecAvg	35.8	1.8	1.6	2.2	3.6	4.2	4.2	3.9	3.9	3.1	2.9	2.6	1.9
	PrecMax	45.4	4.1	5.6	5.0	6.9	7.4	9.9	7.7	9.6	12.6	7.9	5.9	4.9
	PrecMin	23.7	0.4	0.4	0.7	1.5	0.7	0.9	1.2	1.6	0.6	0.5	0.4	0.5
	PrecStd	5.0	1.0	1.1	1.1	1.5	1.6	2.2	1.5	1.6	2.4	1.9	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	61.1	77.0	82.9	90.0	92.9	94.9	93.1	90.8	84.1	69.9	61.2
		MCWB	54.3	53.6	63.7	62.4	69.0	72.3	76.0	78.0	73.1	70.8	58.0	56.7
	2%	DB	49.6	53.2	67.5	77.6	85.5	89.9	90.4	89.6	87.4	78.5	63.7	54.2
		MCWB	45.8	48.4	58.2	60.5	68.1	72.3	75.3	74.9	72.3	64.2	55.7	51.1
	5%	DB	44.0	45.8	61.5	72.5	81.1	86.1	87.7	86.0	82.4	72.8	59.9	48.1
		MCWB	41.0	41.7	54.1	58.7	65.8	70.9	74.5	73.2	68.7	62.2	54.0	45.5
	10%	DB	39.0	40.9	55.8	67.8	76.5	82.2	84.4	82.4	79.2	68.1	55.0	43.9
		MCWB	36.6	37.4	49.3	56.3	64.7	69.2	72.5	71.3	67.2	60.8	49.5	41.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	56.4	64.8	67.2	74.3	78.2	80.3	81.3	77.3	72.3	61.7	57.7
		MCDB	57.2	57.6	74.0	77.4	84.8	87.2	89.0	88.2	85.9	81.2	65.5	60.6
	2%	WB	46.4	49.1	60.2	63.5	70.9	75.4	77.9	78.1	74.2	68.1	57.8	51.3
		MCDB	48.2	53.1	66.0	73.3	80.1	84.0	87.0	85.7	81.9	73.3	62.3	52.9
	5%	WB	41.1	42.5	55.9	60.5	68.4	73.4	76.1	75.8	72.0	65.0	54.1	45.9
		MCDB	44.0	46.0	60.6	69.8	76.8	82.5	84.6	82.7	79.7	70.0	58.1	47.4
	10%	WB	36.9	37.6	49.8	57.4	66.3	71.6	74.5	73.6	69.9	61.4	50.4	41.0
		MCDB	39.2	40.2	55.8	66.4	73.8	79.8	82.2	79.9	76.5	67.2	55.0	43.7
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.1	16.7	20.2	20.5	19.9	19.2	18.8	20.3	18.6	15.7	12.9
		MCDBR	17.7	20.4	24.0	28.3	25.2	23.0	21.9	22.2	23.8	25.2	21.8	17.6
		MCWBR	15.1	16.3	16.1	15.9	12.0	10.9	10.1	10.6	10.9	13.9	15.4	15.6
	5% WB	MCDBR	17.0	20.1	22.0	25.3	21.8	19.4	19.1	18.6	19.7	20.1	19.1	16.5
		MCWBR	15.2	16.9	16.0	16.2	12.6	11.0	10.7	10.7	10.3	13.6	15.4	15.7
Clear-Sky Solar Irradiance	taub	0.296	0.316	0.349	0.402	0.432	0.437	0.451	0.443	0.415	0.359	0.315	0.297	
	taud	2.423	2.372	2.374	2.236	2.208	2.235	2.194	2.209	2.258	2.414	2.489	2.449	
	Ebn at Noon	272	283	285	275	269	267	262	260	259	263	263	261	
	Edh at Noon	25	30	34	43	45	44	45	43	38	29	23	23	
All-Sky Solar Radiation	RadAvg	521	800	1139	1472	1679	1919	1924	1687	1410	931	597	439	
	RadStd	54	106	87	120	139	133	102	97	120	103	57	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page